

The more, the merrier: the role of exogeneity in global state space models

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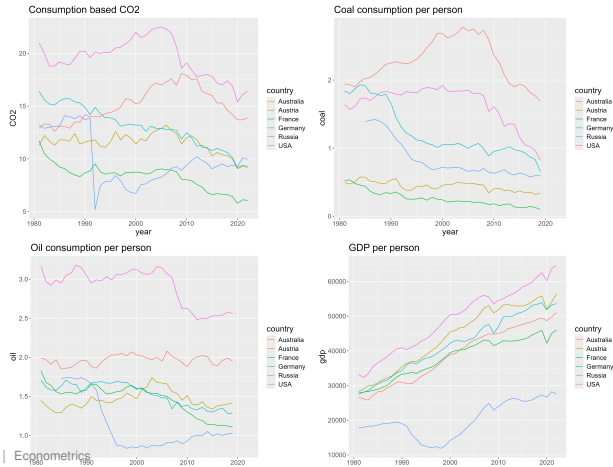
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Introduction

Structured large dimensional data sets

Yearly data from 6 countries/regions on 4 variables ($6 * 4 = 24$) out of a much larger set in Gapminder: CO2, oil, coal consumption per person, gdp per person, yearly data (42 years of obs).



Models for such data

- Autoregressive models (VAR) contain too many parameters. Restrictions are needed.
- Factor models are an option (approximate dynamic factor models).
- Bayesian VARs allow for the imposition of sparsity enhancing priors.
- Reduced rank restriction allows to reduce the dimensionality of the parameter set.
- Matrix Time Series (MaTS) models are another option.
- None of these approaches uses structural information on the relation between the variables that may be available from macro-economics theory, for example.
- In particular, co-integrating relations based on prior knowledge are hard to restrict in these approaches.

The GVAR idea

- Hashem Pesaran proposed to partition the vector $y_t \in \mathbb{R}^S$ of variables into N subsets $y_{i,t} \in \mathbb{R}^{s_i}$, called 'regions' ($\sum_{i=1}^N s_i = S$).
- Then a separate VAR model is estimated for the fewer variables of each of the N regions:

$$y_{i,t} = A_1 y_{i,t-1} + \cdots + A_p y_{i,t-p} + B_0 y_{i,t}^* + \cdots + B_q y_{i,t-q}^* + \varepsilon_{i,t}.$$

- The VAR model contains **exogenous** variables, called 'star' variables.
- The star variables $y_{i,t}^* = W_i' y_t$ are weighted averages of variables from all but the i -th region ($W_{ii} = 0$).
- This allows for the imposition of structural relations in the regional models.
- For example, vector error correction models (VECM) can be used to impose long-run restrictions:

$$\Delta y_{i,t} = \alpha_i \beta_i' \begin{pmatrix} y_{i,t-1} \\ y_{i,t-1}^* \end{pmatrix} + \Gamma_1 \Delta y_{i,t-1} + \cdots + \Gamma_p \Delta y_{i,t-p} + \Theta_0 \Delta y_{i,t}^* + \cdots + \Theta_q \Delta y_{i,t-q}^* + \varepsilon_{i,t}.$$

REMARK: motivation: macro-economic models for different countries.

Key concept: exogeneity

The GVAR idea hinges on the concept of **exogeneity** such that the estimation can be done partitioned into sub-models.

The term **exogeneity** is heavily overloaded with several different notions. We will discuss several concepts.

- For each concept I will investigate the properties of the estimators.
- I will also investigate, what goes wrong, if exogeneity does not hold.
- Finally I will discuss the GLASS framework (extending the GVAR to state space processes), that solves some of the issues in the GVAR.

Exogeneity in cross sectional regressions

Cross sectional case, CLM: strict exogeneity

$$y_i = \beta' x_i + u_i$$

- OLS: $\hat{\beta} = (X'X)^{-1}X'y$.
- Under the CLM: BLUE with distribution $\mathcal{N}(\beta_o, \sigma^2(X'X)^{-1})$ conditional on X .
- For conditional inference we do not need to know the properties of x_i !
- However, some properties depend on x_i : consistency, for example: If $x_i = \frac{1}{i}$ then

$$\sum_{i=1}^N x_i^2 = \sum_{i=1}^N \frac{1}{i^2} \xrightarrow{N \rightarrow \infty} \frac{\pi^2}{6} < \infty \Rightarrow \text{consistency fails!}$$

- For $x_i = i$ on the other hand, $\sqrt{N}(\hat{\beta} - \beta_o) \rightarrow 0$.
- Thus the asymptotics of $\sqrt{N}(\hat{\beta} - \beta_o)$ depend on x_i .

Consequences of violation of strict exogeneity

- If $\mathbb{E}(U|X) \neq 0$, then OLS is biased.
- Alternative estimators must be used such as instrumental variables.
- Thus it is important to detect endogeneity. This can be done using endogeneity tests in the IV literature.
- Asymptotics depend on the properties of x_i even under strict exogeneity.

Time series context

3. Time series context



- In a time series context it is often not possible to condition on all x_s , but only on current x_t .
- Consider as an example a bivariate VAR(1):

$$y_{1,t} = A_{11}y_{1,t-1} + A_{12}y_{2,t-1} + \varepsilon_{1,t},$$

$$y_{2,t} = A_{21}y_{1,t-1} + A_{22}y_{2,t-1} + \varepsilon_{2,t}.$$

- Each of these equations can be estimated using OLS.
- The estimates are consistent and standard normal, if the eigenvalues of the matrix $\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ are all smaller than 1.
- If one only considers the first equation, one obtains a VARX(1) model.
- Structural relations may lead to the addition of a concurrent term:

$$y_{1,t} = A_{11}y_{1,t-1} + A_{12}y_{2,t-1} + D_1y_{2,t} + \nu_{1,t}.$$

Conditions allowing separation

Assumption (weak exogeneity, Engle et al. (1983))

The process $y_{2,t}$ in the VARX(1) model for $y_{1,t}$ is *weakly exogenous for the parameter vector ϕ_1* if the likelihood for the t -th observation of the joint process (conditional on the joint past) can be factored as:

$$L_j(\phi_1, \phi_2; y_{1,t}, y_{2,t} | Y_{t-1}) = L_1(\phi_1; y_{1,t} | y_{2,t}, Y_{t-1}) L_2(\phi_2; y_{2,t} | Y_{t-1}).$$

where $Y_{t-1} = (y_{1,t-j}, y_{2,t-j}, j = 1, \dots, t)$ denotes the information from prior to t .

In the case of Gaussian distributed processes in

$$y_{1,t} = A_{11}y_{1,t-1} + A_{12}y_{2,t-1} + D_1y_{2,t} + \nu_{1,t}.$$

- $y_{2,t}$ and $\nu_{1,t}$ need to be uncorrelated,
- Then $\phi_1 = (A_{11}, A_{12}, D_1, \Omega_1)$ and $\phi_2 = (A_{21}, A_{22}, \Omega_2)$ do not share common parameters.

Inference under weak exogeneity

$$\log L_j(\phi_1, \phi_2; y_{1,t}, y_{2,t} | Y_{t-1}) = \log L_1(\phi_1; y_{1,t} | y_{2,t}, Y_{t-1}) + \log L_2(\phi_2; y_{2,t} | Y_{t-1}).$$

- If the likelihood can be factored, the log likelihood is the sum of two terms.
- Since the second term does not depend on ϕ_1 ,
optimal estimation and inference on ϕ_1 can be done based only on the first part of the likelihood.
- In the Gaussian case this amounts to estimating only the equation for $y_{1,t}$.
- Hannan and Deistler (1988) show that for VARMAX models for $y_{1,t}$ conditional on $y_{2,t}$ the asymptotically normal distribution for the MLE for ϕ_1 only depends on the covariance sequence for $y_{2,t}$, if this process is stationary.
- It is different, if the process is integrated, however.

Testing weak exogeneity in the VAR(1) setting

Consider the model.

$$\begin{array}{llll}
 \text{(marginal)} & y_{2,t} & = & A_{21}y_{1,t-1} + A_{22}y_{2,t-1} + \varepsilon_{2,t} \quad (I) \\
 \text{(conditional)} & y_{1,t} & = & A_{11}y_{1,t-1} + A_{12}y_{2,t-1} + D_1 y_{2,t} + \nu_{1,t} \quad (II) \\
 \Rightarrow & y_{1,t} & = & (A_{11} + D_1 A_{21})y_{1,t-1} + (A_{12} + D_1 A_{22})y_{2,t-1} + \tilde{\varepsilon}_{1,t} \quad (III)
 \end{array}$$

- We can estimate the joint VAR(1) (I+III) also if weak exogeneity does not hold (and thus $\varepsilon_{2,t}$ and $\nu_{1,t}$ are correlated) $\Rightarrow \varepsilon_{t,2}$

- Standard endogeneity test estimates:

$$y_{1,t} = A_{11}y_{1,t-1} + A_{12}y_{2,t-1} + D_1 y_{2,t} + D_1 \varepsilon_{2,t} + \nu_{1,t} \quad (II')$$

- If we find no evidence for violation of weak exogeneity ($D_1 \neq 0$), we may use the conditional model (II).
- If we find evidence, then we have to use the joint model (I+III): This implies that we need a model for $y_{2,t}$.

From weak to strong exogeneity

Dynamics under weak exogeneity

Example:

$$y_{1,t} = \rho y_{1,t-1} + \rho y_{2,t-1} + \varepsilon_{1,t},$$

$$y_{2,t} = \rho y_{1,t-1} + \rho y_{2,t-1} + \varepsilon_{2,t}$$

- The eigenvalues of $A = \rho \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ equal 0 and 2ρ .
- For $|\rho| < 0.5$ we get a stable matrix.
- For $\rho = 0.5$ the process $y_{1,t} + y_{2,t}$ is a random walk.
- For $|\rho| > 0.5$ it is explosive.

The conditional model equals:

$$y_{1,t} = \rho y_{1,t-1} + A_{12} y_{2,t-1} + \varepsilon_{1,t}.$$

- Thus the stability condition equals $|\rho| < 0.5$.
- If $y_{2,t} = A_{22} y_{2,t-1} + \varepsilon_{2,t}$ for $|A_{22}| < 1$ the stability condition is $|\rho| < 1$.
- Thus the properties of the process defined in the conditional model depend on the properties of the marginal model in general.

Forecasting

Consider forming the two step forecasts:

$$\begin{aligned}
 y_{1,T+2} &= \rho y_{1,T+1} + \rho y_{2,T+1} + \varepsilon_{1,T+2} \\
 &= \rho(\rho y_{1,T} + \rho y_{2,T} + \varepsilon_{1,T+1}) + \rho \hat{y}_{2,T+1|T} + \rho(y_{2,T+1} - \hat{y}_{2,T+1|T}) + \varepsilon_{1,T+2} \\
 &= \underbrace{\rho^2 y_{1,T} + \rho^2 y_{2,T} + \rho \hat{y}_{2,T+1|T}}_{\text{prediction}} + \underbrace{\rho \varepsilon_{1,T+1} + \rho(y_{2,T+1} - \hat{y}_{2,T+1|T})}_{\text{prediction error}} + \varepsilon_{1,T+2}.
 \end{aligned}$$

- involve predictions for $y_{2,T+1}$.
- use the marginal model.
- predictions for $y_{2,T+j}$ require the previous predictions for $y_{1,T+j-i}$.
- Thus we need both models to obtain predictions. We cannot set the predictions for $y_{2,T+j}$ to arbitrary values (policy analysis!).

Granger causality

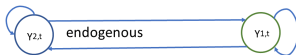
The problem: feedback from $y_{1,t}$ onto $y_{2,t}$: $y_{2,t}$ depends on the past of $y_{1,t}$.

Definition (Granger causality, somewhat sloppy)

$(y_{1,t})$ does not Granger cause $(y_{2,t})$, if adding $y_{1,t-j}, j = 1, 2, \dots$ to the linear model explaining $y_{2,t}$ does not improve the predictions.

$$y_{2,t} = \sum_{j=1}^{\infty} A_{22,j} y_{2,t-j} + \sum_{j=1}^{\infty} A_{21,j} y_{1,t-j} + \varepsilon_{2,t}.$$

For Granger non-causality: $A_{21,j} = 0$.



Assumption (strong exogeneity, Engle et al. (1983))

The process $(y_{2,t})$ is said to be strongly exogenous for estimating ϕ_1 , if it is weakly exogenous for ϕ_1 and additionally, $(y_{1,t})$ does not Granger cause $(y_{2,t})$.

- To test this, we must ascertain both parts.
- Granger non-causality is a restriction on the parameter set.
- In the VAR(1) setting this relates to $A_{21} = 0$.
- In the GVAR, the joint process $(y'_{i,t}, (y'_{i,t})')'$ is not a VAR, but can only be approximated by a VAR.
- This may require a large lag length and reduce power of the test.
- For state space models this is different: In that case the joint process again is a state space process.

The GLASS model

Regional models

$$\begin{aligned}y_{i,t} &= C_i x_{i,t} + D_i y_{i,t}^* + \varepsilon_{i,t} \\x_{i,t+1} &= \underline{A}_i x_{i,t} + B_i y_{i,t}^* + K_i y_{i,t}.\end{aligned}$$

- The vector of regional variables $y_{i,t} \in \mathbb{R}^{s_i}$ is given by a state space system with additional variables $y_{i,t}^*$ appearing in the state and the measurement equation.
- $x_{i,t} \in \mathbb{R}^{n_i}$ are the latent states for region i .
- Star variables: $y_{i,t}^* = \sum_{j=1}^N W_{i,j} y_{j,t}$, $W_{i,i} = 0$.
- The weights $W_{i,j}$ are typically defined using trade relations (see Chudik and Pesaran, 2016).

Joint model

$$\underbrace{\begin{pmatrix} y_{1,t} \\ \vdots \\ y_{N,t} \end{pmatrix}}_{y_t} = \underbrace{\begin{pmatrix} C_1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & C_N \end{pmatrix}}_{C_{\odot}} x_t + D_{\odot} y_t^* + \varepsilon_t,$$

$$\underbrace{\begin{pmatrix} x_{1,t+1} \\ \vdots \\ x_{N,t+1} \end{pmatrix}}_{x_{t+1}} = \underline{A}_{\odot} x_t + B_{\odot} y_t^* + K_{\odot} y_t, \quad y_t^* = \begin{pmatrix} W'_1 \\ \vdots \\ W'_N \end{pmatrix} y_t = W y_t$$

$$\Rightarrow (I - D_{\odot} W) y_t = C_{\odot} x_t + \varepsilon_t, \quad x_{t+1} = \underline{A}_{\odot} x_t + (B_{\odot} W + K_{\odot}) y_t.$$

$$y_t = (I - D_{\odot} W)^{-1} C_{\odot} x_t + (I - D_{\odot} W)^{-1} \varepsilon_t.$$

Properties of joint model

$$\begin{aligned}y_t &= (I - D_{\odot} W)^{-1} C_{\odot} x_t + (I - D_{\odot} W)^{-1} \varepsilon_t = \tilde{C} x_t + \tilde{\varepsilon}_t, \\x_{t+1} &= \underline{A}_{\odot} x_t + (B_{\odot} W + K_{\odot}) y_t, \\x_{t+1} &= \underline{A}_{\odot} x_t + (B_{\odot} W + K_{\odot})(\tilde{C} x_t + \tilde{\varepsilon}_t) \\&= (\underline{A}_{\odot} + (B_{\odot} W + K_{\odot}) \tilde{C}) x_t + (B_{\odot} W + K_{\odot}) \tilde{\varepsilon}_t\end{aligned}$$

- Invertibility (long VAR approximation) governed by $\underline{A}_{\odot}$: controlled in the regional models.
- Stability: eigenvalues of $(\underline{A}_{\odot} + (B_{\odot} W + K_{\odot})(I - D_{\odot} W)^{-1} C_{\odot})$ in general depend on all regional models.

(I) Weak exogeneity

- The correlation between $y_{i,t}^*$ and $\varepsilon_{i,t}$ can be calculated as:

$$0 = W_i'(I - D_{\odot} W)^{-1} \Omega E_i, \quad \text{Var}(\varepsilon_t, \varepsilon_t) = \Omega,$$

$$E_i' = (0 \quad \dots \quad 0 \quad I \quad 0 \quad \dots \quad 0).$$

- Typically block diagonality of Ω is assumed:
 $\Omega = \Omega_{\odot} \Rightarrow \Omega E_i = E_i \Omega_i \Rightarrow 0 = W_i'(I - D_{\odot} W)^{-1} E_i \Omega_i.$
- This in general involves all regional models.
- An exception occurs for $D_{\odot} = 0$. Then block diagonality is sufficient for weak exogeneity.

THUS: In general weak exogeneity in the GLASS involves all regional models, but can be tested in each regional model separately.

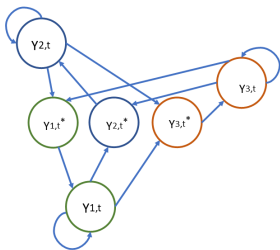
Consequences

- Test weak exogeneity.
- If it is plausible: continue.
- Restrictions such that weak exogeneity holds cannot be imposed $\Rightarrow$ loss of efficiency.
- If it is not plausible: use reduced form model, alternative estimators.

(II) Granger causality: feedback paths

$$y_{i,t}^* = W_i' y_t = W_i' (I - D_{\odot} W)^{-1} (C_{\odot} x_t + \varepsilon_t),$$

$$y_{i,t} = C_i x_{i,t} + D_i y_{i,t}^* + \varepsilon_{i,t}.$$



Theorem

In the GLASS model the conditions for all star variables $y_{i,t}^$ not to be Granger caused by the regional variables $y_{i,t}$ for $i = 1, \dots, N$ hold almost nowhere (the set of parameters, where the conditions hold is a Lebesgue null set).*

Thus: To ensure Granger non-causality, we need to impose restrictions on all regional models simultaneously.

Example: Two region VAR(1)

$$\begin{aligned} y_{1,t} &= A_{11}y_{1,t-1} + \tilde{A}_{12}W_{12}y_{2,t-1} + \varepsilon_{1,t}, \\ y_{2,t} &= \tilde{A}_{21}W_{21}y_{1,t-1} + A_{22}y_{2,t-1} + \varepsilon_{2,t}. \end{aligned}$$

Strong exogeneity:

- $y_{1,t}$ may not Granger cause $y_{1,t}^* = W_{12}y_{2,t}$:

$$W_{12}y_{2,t} = W_{12}\tilde{A}_{21}W_{21}y_{1,t-1} + W_{12}A_{22}y_{2,t-1} + W_{12}\varepsilon_{2,t}.$$

- $y_{2,t}$ may not Granger cause $y_{2,t}^* = W_{21}y_{1,t}$.

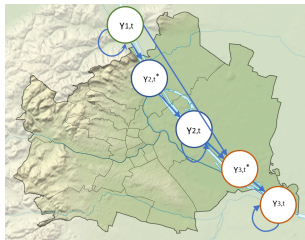
$$W_{21}y_{1,t} = W_{21}A_{11}y_{1,t-1} + W_{21}\tilde{A}_{12}W_{12}y_{2,t-1} + W_{21}\varepsilon_{1,t}.$$

Consequences

- The restrictions are necessary for strong exogeneity.
- If one ignores them, it is not guaranteed that strong exogeneity holds.
- Then the properties such as stability or the existence of unit roots cannot be controlled in the regional models alone.
- Since the restrictions involves all regional models, they cannot be imposed within the regional model alone.
- In the stationary case, one may ignore them for the estimation of the regional model and hope that the estimated joint model is stationary.
- If long-run relations are of interest, this approach fails.



Blt structure



Special case:

- block lower triangular (blt) structure for W : regions are ordered, influence only runs 'downstream'.
- Then $D_{\odot} W$ and $(I - D_{\odot} W)$ are blt.
- So are $(I - D_{\odot} W)^{-1} C_{\odot}$ and $(B_{\odot} W + K_{\odot})$ and $A = \underline{A}_{\odot} + K_{\odot} C_{\odot} + \tilde{A}$ (b.l.t. with zero diagonal)

Theorem

In the GLASS model, if the weight matrix W is block lower triangular and if the matrix $\Omega = \Omega_{\odot}$ is block diagonal, then

- *in all models the star variables are strongly exogenous,*
- *the regional variables $y_{i,t}$ do not Granger cause $y_{i,t}^*$.*

Grafik Wien: Von NordNordWest - Eigenes Werk using SRTM30 v.2 data, CC BY-SA 3.0 de,
<https://commons.wikimedia.org/w/index.php?curid=23164164>

Working with the blt structure

Sequence of modeling

Since in the blt structure, we have an ordering of the regions, we model sequentially:

- Estimate a state space model for the first region.
- Estimate recursively: the star variables $y_{i,t}^* = \sum_{j=1}^{i-1} W_{i,j} y_{j,t}$ depend on the previous regions (for which we already have a model).
- Given the star variables, estimate the regional model for $y_{i,t}$.
- Test strong exogeneity. If it holds, continue.
- If strong exogeneity does not hold use the joint model for $(y'_{i,t}, (y_{i,t}^*)')'$.
- If all N regional models are obtained, stack the models to obtain the joint model.

6. Working with the blt structure



Joint model in the i -th region:

$$\begin{pmatrix} y_{i,t}^* \\ y_{i,t} \end{pmatrix} = \begin{pmatrix} W_i' \\ E_i' \end{pmatrix} y_t = \tilde{C}_i \tilde{x}_{i,t} + \tilde{\varepsilon}_{i,t}, \tilde{x}_{i,t+1} = \tilde{A}_i \tilde{x}_{i,t} + \tilde{K}_i \tilde{\varepsilon}_{i,t}.$$

Marginal and regional model:

$$\begin{array}{rcl} (marginal) y_{i,t}^* & & = C_i^* x_{i,t}^* + \varepsilon_{i,t}^*, \\ x_{i,t+1}^* & & = \underline{A}_i^* x_{i,t}^* + K_i^* y_{i,t}^*, \\ \hline (regional) y_{i,t} & & = C_i x_{i,t} + D_i y_{i,t}^* + \varepsilon_{i,t}, \\ x_{i,t+1} & & = A_i x_{i,t} + B_i y_{i,t}^* + K_i \varepsilon_{i,t}. \end{array}$$

- (I) In the regional model, we can test for weak exogeneity by adding $\varepsilon_{i,t}^*$.
- (II) $y_{i,t}$ does not Granger cause $y_{i,t}^*$, if the marginal and regional model combined are not significantly worse (in terms of likelihood) than the joint model.

In the blt structure, the stability and invertibility properties of the joint model can be controlled using the regional models.

Reactions to failures of exogeneity

(I) Star vars are not weakly exogenous:

- use the joint model, not marginal + regional.
- use the reduced form (no concurrent variables in the observation equation)

(II) $y_{i,t}$ Granger causes $y_{i,t}^*$:

- use the joint model
- provides an indication that the blt structure does not hold, regions should be redefined.
- information travels 'upstream'.

Conclusions

- The GLASS model extends the GVAR by using a state space model, where the GVAR uses a VAR.
- Within the GLASS model the various exogeneity conditions do not necessarily hold.
- In some sense it is unlikely that they hold.
- The conditions typically involve all regional models jointly.
- In each of the regional models, the conditions can be tested for.
- Depending on the outcome of the tests, one may either use exogeneity or has to estimate the joint model.
- The blt structure is one setting, in which the exogeneity conditions hold automatically for all parameter values (hence no restrictions must be taken into account).
- Algorithms to group variables such that the blt structure holds (resp. reactions to rejections of Granger non-causality tests), are topic of future research.

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- Engle, R.F., Hendry, D.F., Richard, J.F., 1983. Exogeneity. *Econometrica*, 277–304.
- Hannan, E., Deistler, M., 1988. The Statistical Theory of Linear Systems. Wiley.

Thank you for your attention!

Definition

- Integration concerns random walk like behaviour.
- This corresponds to unit eigenvalues of the matrix A_i in the state space formulation:

$$y_{i,t} = (C_{i,1} \quad C_{i,\bullet}) x_{i,t} + D_i y_{i,t}^* + \varepsilon_{i,t},$$

$$\begin{pmatrix} x_{i,t+1,i} \\ x_{i,t+1,\bullet} \end{pmatrix} = \begin{pmatrix} I_c & 0 \\ 0 & A_{i,\bullet} \end{pmatrix} x_{i,t} + \begin{pmatrix} B_{i,1} \\ B_{i,\bullet} \end{pmatrix} y_{i,t}^* + \begin{pmatrix} K_{i,1} \\ K_{i,\bullet} \end{pmatrix} \varepsilon_{i,t}$$

- For weak exogeneity we need a split of the parameters.
- If $y_{i,t}^*$ is integrated and $B_{i,1} \neq 0$, then there is a chance, that $y_{i,t}$ is doubly integrated. Thus one needs to restrict the parameter set.
- For weak exogeneity, we require $B_{i,1} = 0$.
- Typically $y_{i,t}^* = C_{i,1}^* x_{i,t,1}^* + v_{i,t}^*$, where $x_{i,t,1}^*$ contains the common trends driving the process, $v_{i,t}^*$ denotes all stationary summands.
- The co-dimension of $C_{i,1}^*$ (the dimension of the left kernel $\beta' C_{i,1}^* = 0$) is called co-integrating rank.
- The properties of the regional process $y_{i,t}$ depend on this cointegrating rank.

Testing for the co-integrating rank

- The Johansen trace test uses asymptotic distributions to obtain critical values.
- The limiting distribution depends on properties of the exogenous variables $y_{i,t}^*$, above all on the cointegrating rank.
- The weak exogeneity assumption $B_{i,1} = 0$ is a restriction on the parameters and can be tested.
- Under this assumption the determination of the cointegrating rank using the Johansen trace test still depends on the cointegrating rank of the star variables, but does not contain any other nuisance parameters.

